Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/20/2006
PI Name:	Buckley, Steven G. Ph.D.		
Project Title:	Material Properties Governing Co-Current Flame Spread in Micro-Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Diego	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	07/01/2003	**End Date:**	04/07/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NNC04GB11G		
Performance Goal No.:			
Performance Goal Text:			

The current flammability requirements for all materials to be used in space vehicles (NASA specifications) are given by the 'Flammability, Odor, Offgassing, and Compatibility Requirements and Test Procedures for Materials in Environments that Support Combustion' document. This document specifies two tests that need to be performed before a material is qualified to be used in a space vehicle, the 'Upward Flame Propagation Test' (Test 1) and the 'Heat and Visible Smoke Release Rates Test' (Test 2). These two tests are expected to properly assess the flammability of a material in micro-gravity conditions. These methods attempt to provide a worst case scenario (Test 1) and a measure of the heat release (Test 2), and consequently, the 'damage potential' of a fire. A detailed description of these test methods is provided in NASA-NHB 8060.1 and an extensive list of the materials that have been tested is provided in the 'Materials Selection List for Space Hardware Systems.'
Recent work, as reported in the proposal for this project and the publications from this research, suggest that current methods of assessing material flamability may be improved. In essence, no single test current test method is capable of

Task Description:	providing enough information to fully describe fire growth. Analytical and model-based solutions are sought to support the wealth of earth-gravity flame spread data that is available, to provide a bridge between current fundamental studies and critical fire safety needs. Experiments with advanced diagnostics are performed with PMMA, a well-characterized material that can be readily compared with theory. The goal of this work is thus to provide an improved methodology for assessment of material fire safety in micro-gravity. The objective this work is to provide an improved analytical framework for assessment of material fire safety in micro-gravity environments, based on current fundamental understanding of flame spread. Such a framework would allow the extrapolation of a wealth of earth-gravity flammability data to micro-gravity and allow connection between fundamental knowledge and vital practical safety requirements. Current and planned human space missions are of such duration that the risk of fire could be significant, and the impact of a potential fire could be catastrophic. Improved understanding of material flammability is crucial to minimizing the fire risk. This project will contribute an analytical and model-based framework to better understand and predict material flammability at normal and micro-gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report for this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)